Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	19 January 2026
Team ID	
Project Name	MediGuard – Health Insurance Fraud Detection project
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Interface Setup	USN-1	As a user, I can open the MediGuard web interface to submit insurance claim data for fraud detection.	2	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-1	Claim Input Form	USN-2	As a user, I can enter claim details such as incident type, claim amounts, witnesses, and injuries.	2	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-1	Data Validation	USN-3	As a user, I want the system to validate claim input fields before submission.	1	Medium	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-2	Data Processing	USN-4	As a system, I preprocess claim data using feature engineering and scaling before prediction.	3	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-2	Machine Learning Model	USN-5	As a system, I train ML models such as Decision Tree and Random Forest to detect fraud patterns.	3	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-2	Model Integration	USN-6	As a developer, I integrate trained models into the Flask backend API for predictions.	2	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya

Sprint-3	Fraud Prediction API	USN-7	As a user, I can submit claim data and receive fraud prediction results through the API.	3	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-3	Ensemble Prediction	USN-8	As a system, I combine predictions from multiple ML models to improve accuracy.	2	Medium	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-3	AI Investigator Report	USN-9	As a user, I receive an AI-generated explanation for the fraud prediction.	2	Medium	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-4	Results Dashboard	USN-10	As a user, I can view fraud probability, model predictions, and verdict on the dashboard.	3	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-4	Model Accuracy Comparison	USN-11	As a user, I can compare training and testing accuracy of all ML models.	2	Medium	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya
Sprint-4	Final Testing & Deployment	USN-12	As a developer, I test the system and deploy the MediGuard fraud detection application.	2	High	Maroof Gadiwale, Shankar Kamble, Shraddhesh Tamhane, Rajveersingh Sisodiya

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date
Sprint-1	20	10 Days	19 Jan 2025	29 Jan 2025	20	29 Jan 2025
Sprint-2	20	10 Days	30 Jan 2025	09 Feb 2025	20	09 Feb 2025
Sprint-3	20	10 Days	10 Feb 2025	20 Feb 2025	20	20 Feb 2025
Sprint-4	20	10 Days	21 Feb 2025	03 Mar 2025	20	03 Mar 2025
Final Integration	10	5 Days	04 Mar 2025	08 Mar 2025	10	08 Mar 2025

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum.